

Supplementary Results for HISS

Submission to VLDB 2027

Table 1: Performance Comparison on ISCX-VPN-Service with ET-BERT as Backbone.

Dataset	ISCX-VPN-Service							
Method	AC	PR	RC	F1	AC _{0.9}	PR _{0.9}	RC _{0.9}	F1 _{0.9}
ERM	98.87 $\pm$ 0.18	98.87 $\pm$ 0.20	98.87 $\pm$ 0.18	98.87 $\pm$ 0.18	88.65 $\pm$ 1.75	79.22 $\pm$ 8.97	80.20 $\pm$ 4.22	79.34 $\pm$ 5.88
Random	98.92 $\pm$ 0.04	98.94 $\pm$ 0.05	98.92 $\pm$ 0.04	98.92 $\pm$ 0.04	89.15 $\pm$ 0.42	80.62 $\pm$ 7.91	77.49 $\pm$ 6.59	78.40 $\pm$ 6.72
MC-CVaR	98.93 $\pm$ 0.07	98.94 $\pm$ 0.07	98.93 $\pm$ 0.07	98.93 $\pm$ 0.07	89.32 $\pm$ 0.70	80.96 $\pm$ 3.17	85.57 $\pm$ 0.12	81.81 $\pm$ 1.43
OHTM	98.93 $\pm$ 0.23	98.95 $\pm$ 0.23	98.93 $\pm$ 0.23	98.93 $\pm$ 0.23	89.32 $\pm$ 2.30	87.90 $\pm$ 7.06	84.09 $\pm$ 5.82	83.69 $\pm$ 6.01
Focal	98.92 $\pm$ 0.27	98.93 $\pm$ 0.28	98.92 $\pm$ 0.27	98.92 $\pm$ 0.27	89.15 $\pm$ 2.71	89.19 $\pm$ 9.51	86.84 $\pm$ 4.17	86.75 $\pm$ 5.79
GroupDRO	98.97 $\pm$ 0.44	98.97 $\pm$ 0.45	98.97 $\pm$ 0.44	98.97 $\pm$ 0.45	89.65 $\pm$ 4.42	90.69 $\pm$ 9.37	83.36 $\pm$ 5.88	85.71 $\pm$ 6.68
TDRO	98.95 $\pm$ 0.19	98.96 $\pm$ 0.21	98.95 $\pm$ 0.19	98.95 $\pm$ 0.20	89.48 $\pm$ 1.92	89.09 $\pm$ 8.62	85.62 $\pm$ 4.16	86.16 $\pm$ 5.79
HISS w/o penalty	98.93 $\pm$ 0.11	98.95 $\pm$ 0.11	98.93 $\pm$ 0.11	98.93 $\pm$ 0.11	89.32 $\pm$ 1.09	88.42 $\pm$ 2.38	89.33 $\pm$ 0.69	<u>87.67</u> $\pm$ 0.97
HISS	99.12 $\pm$ 0.10	99.13 $\pm$ 0.10	99.12 $\pm$ 0.10	99.12 $\pm$ 0.10	91.15 $\pm$ 1.00	94.10 $\pm$ 2.12	<u>87.36</u> $\pm$ 1.94	89.47 $\pm$ 1.78

Note: Results are percentages and reported as mean $\pm$ std. Best and second-best mean values are in bold and underlined, respectively.

Table 2: Performance Comparison on ISCX-VPN-App with ET-BERT as Backbone.

Dataset	ISCX-VPN-App							
Method	AC	PR	RC	F1	AC _{0.9}	PR _{0.9}	RC _{0.9}	F1 _{0.9}
ERM	99.61 $\pm$ 0.10	99.27 $\pm$ 0.26	99.37 $\pm$ 0.12	99.32 $\pm$ 0.19	96.11 $\pm$ 0.98	90.25 $\pm$ 6.44	87.37 $\pm$ 7.23	88.20 $\pm$ 6.72
Random	99.52 $\pm$ 0.11	99.14 $\pm$ 0.26	99.25 $\pm$ 0.15	99.19 $\pm$ 0.20	95.20 $\pm$ 1.11	82.24 $\pm$ 6.04	77.33 $\pm$ 5.99	77.78 $\pm$ 5.63
MC-CVaR	99.61 $\pm$ 0.03	99.46 $\pm$ 0.13	99.46 $\pm$ 0.16	99.46 $\pm$ 0.09	96.11 $\pm$ 0.27	82.43 $\pm$ 1.41	81.14 $\pm$ 0.72	81.10 $\pm$ 0.46
OHTM	99.51 $\pm$ 0.09	99.18 $\pm$ 0.18	99.40 $\pm$ 0.11	99.28 $\pm$ 0.14	95.07 $\pm$ 0.91	78.06 $\pm$ 5.12	74.86 $\pm$ 5.01	75.27 $\pm$ 4.60
Focal	99.65 $\pm$ 0.17	99.35 $\pm$ 0.36	99.47 $\pm$ 0.25	99.41 $\pm$ 0.31	96.50 $\pm$ 1.66	97.47 $\pm$ 4.36	<u>91.72</u> $\pm$ 5.51	93.56 $\pm$ 5.16
GroupDRO	99.66 $\pm$ 0.12	99.33 $\pm$ 0.27	<u>99.54</u> $\pm$ 0.14	99.43 $\pm$ 0.17	96.63 $\pm$ 1.25	86.37 $\pm$ 8.53	82.01 $\pm$ 8.13	83.14 $\pm$ 8.51
TDRO	<u>99.69</u> $\pm$ 0.09	99.37 $\pm$ 0.31	99.47 $\pm$ 0.13	99.41 $\pm$ 0.19	<u>96.89</u> $\pm$ 0.91	91.02 $\pm$ 5.63	87.45 $\pm$ 5.53	88.65 $\pm$ 5.74
HISS w/o penalty	99.70 $\pm$ 0.15	<u>99.38</u> $\pm$ 0.16	99.51 $\pm$ 0.20	99.44 $\pm$ 0.18	97.02 $\pm$ 1.46	92.53 $\pm$ 2.44	83.50 $\pm$ 3.40	86.15 $\pm$ 2.93
HISS	99.68 $\pm$ 0.09	99.36 $\pm$ 0.25	99.55 $\pm$ 0.17	<u>99.45</u> $\pm$ 0.21	96.76 $\pm$ 0.87	<u>95.86</u> $\pm$ 2.64	91.77 $\pm$ 2.97	<u>93.30</u> $\pm$ 2.81

Note: Results are percentages and reported as mean $\pm$ std. Best and second-best mean values are in bold and underlined, respectively.

Table 3: Performance Comparison on USTC-TFC with ET-BERT as Backbone.

Dataset	USTC-TFC							
Method	AC	PR	RC	F1	AC _{0.9}	PR _{0.9}	RC _{0.9}	F1 _{0.9}
ERM	99.05 $\pm$ 0.07	99.07 $\pm$ 0.07	99.07 $\pm$ 0.08	99.07 $\pm$ 0.07	90.53 $\pm$ 0.71	51.97 $\pm$ 1.42	52.93 $\pm$ 1.20	52.35 $\pm$ 0.86
Random	98.97 $\pm$ 0.13	98.99 $\pm$ 0.11	98.99 $\pm$ 0.14	98.99 $\pm$ 0.14	89.70 $\pm$ 1.34	48.62 $\pm$ 1.72	50.71 $\pm$ 1.97	48.76 $\pm$ 1.21
MC-CVaR	99.18 $\pm$ 0.21	99.21 $\pm$ 0.17	99.19 $\pm$ 0.20	99.18 $\pm$ 0.20	91.76 $\pm$ 2.06	47.38 $\pm$ 1.12	48.67 $\pm$ 2.58	46.56 $\pm$ 1.54
OHTM	98.98 $\pm$ 0.24	99.00 $\pm$ 0.23	99.00 $\pm$ 0.24	99.00 $\pm$ 0.24	89.80 $\pm$ 2.44	45.08 $\pm$ 3.73	44.31 $\pm$ 2.19	44.41 $\pm$ 2.40
Focal	<u>99.12</u> $\pm$ 0.14	<u>99.14</u> $\pm$ 0.13	<u>99.14</u> $\pm$ 0.14	<u>99.14</u> $\pm$ 0.14	<u>91.25</u> $\pm$ 1.40	66.52 $\pm$ 2.57	66.49 $\pm$ 3.70	66.49 $\pm$ 4.03
GroupDRO	98.83 $\pm$ 0.04	98.85 $\pm$ 0.01	98.85 $\pm$ 0.02	98.85 $\pm$ 0.02	88.26 $\pm$ 0.36	66.63 $\pm$ 5.12	66.74 $\pm$ 4.23	66.65 $\pm$ 4.57
TDRO	99.08 $\pm$ 0.03	99.10 $\pm$ 0.01	99.10 $\pm$ 0.03	99.10 $\pm$ 0.03	90.83 $\pm$ 0.26	44.29 $\pm$ 2.65	46.03 $\pm$ 1.71	44.82 $\pm$ 1.43
HISS w/o penalty	99.04 $\pm$ 0.12	99.07 $\pm$ 0.10	99.06 $\pm$ 0.11	99.06 $\pm$ 0.11	90.42 $\pm$ 1.19	<u>71.61</u> $\pm$ 3.53	<u>71.31</u> $\pm$ 3.43	<u>71.43</u> $\pm$ 3.39
HISS	98.97 $\pm$ 0.04	99.01 $\pm$ 0.04	98.99 $\pm$ 0.04	98.99 $\pm$ 0.04	89.70 $\pm$ 0.44	74.07 $\pm$ 6.46	75.39 $\pm$ 6.61	74.12 $\pm$ 6.52

Note: Results are percentages and reported as mean $\pm$ std. Best and second-best mean values are in bold and underlined, respectively.

Table 4: Performance Comparison on CSTNET-TLS 1.3 with ET-BERT as Backbone.

Dataset	CSTNET-TLS 1.3							
Method	AC	PR	RC	F1	AC _{0.9}	PR _{0.9}	RC _{0.9}	F1 _{0.9}
ERM	95.51 $\pm$ 0.10	95.56 $\pm$ 0.10	95.56 $\pm$ 0.10	95.54 $\pm$ 0.10	55.11 $\pm$ 0.97	56.43 $\pm$ 1.12	54.41 $\pm$ 0.71	53.72 $\pm$ 0.83
Random	95.46 $\pm$ 0.16	95.58 $\pm$ 0.17	95.49 $\pm$ 0.16	95.50 $\pm$ 0.17	54.58 $\pm$ 1.61	58.03 $\pm$ 0.96	53.21 $\pm$ 1.69	53.35 $\pm$ 1.86
MC-CVaR	96.07 $\pm$ 0.01	96.11 $\pm$ 0.03	96.11 $\pm$ 0.01	96.09 $\pm$ 0.02	60.67 $\pm$ 0.07	61.47 $\pm$ 0.38	<u>60.27</u> $\pm$ 0.49	58.97 $\pm$ 0.12
OHTM	95.31 $\pm$ 0.13	95.42 $\pm$ 0.13	95.35 $\pm$ 0.14	95.36 $\pm$ 0.13	53.14 $\pm$ 1.29	55.51 $\pm$ 1.78	51.54 $\pm$ 1.09	51.16 $\pm$ 1.25
Focal	95.24 $\pm$ 0.12	95.29 $\pm$ 0.14	95.30 $\pm$ 0.12	95.27 $\pm$ 0.13	52.42 $\pm$ 1.17	52.87 $\pm$ 1.89	50.34 $\pm$ 0.83	49.99 $\pm$ 1.16
GroupDRO	88.86 $\pm$ 0.17	89.12 $\pm$ 0.16	88.87 $\pm$ 0.17	88.93 $\pm$ 0.16	13.19 $\pm$ 1.66	15.05 $\pm$ 1.98	12.34 $\pm$ 1.44	12.40 $\pm$ 1.65
TDRO	94.50 $\pm$ 0.10	94.56 $\pm$ 0.11	94.55 $\pm$ 0.11	94.54 $\pm$ 0.11	44.97 $\pm$ 1.00	41.25 $\pm$ 1.48	40.18 $\pm$ 1.18	39.49 $\pm$ 1.45
HISS w/o penalty	95.98 $\pm$ 0.14	<u>96.06</u> $\pm$ 0.13	96.02 $\pm$ 0.13	96.01 $\pm$ 0.14	59.82 $\pm$ 1.41	61.66 $\pm$ 1.30	60.08 $\pm$ 1.30	<u>59.07</u> $\pm$ 1.22
HISS	<u>96.03</u> $\pm$ 0.10	<u>96.06</u> $\pm$ 0.10	<u>96.10</u> $\pm$ 0.11	<u>96.05</u> $\pm$ 0.11	<u>60.31</u> $\pm$ 0.98	<u>61.57</u> $\pm$ 0.64	60.98 $\pm$ 0.89	59.69 $\pm$ 0.90

Note: Results are percentages and reported as mean $\pm$ std. Best and second-best mean values are in bold and underlined, respectively.